REWARDMARK: Ownership Watermarking for RLHF Reward Models

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Abstract

A high-quality RLHF reward model (RM) is now a tradable artifact: `Skywork-Reward`, `ArmoRM`, and `Llama-3.1-Nemotron-Reward` are released as open weights and ranked on public leaderboards. Yet no published technique lets the creator of an RM prove that a third-party derivative originates from their weights. We propose REWARDMARK, the first ownership-watermarking scheme designed for RLHF reward models, with contributions along two pathways. **Pathway A (RM-direct verification)** is fully solved: a hidden $(\mathcal{T}, \sigma)$ trigger injected via a composite Bradley-Terry/margin objective produces a watermark detectable on the suspect RM (VERIFY-A: paired Wilcoxon $p < 5.8 \times 10^{-10}$, median score margin $+3.76$ at 7B / $+2.63$ at 3B on $K{=}50$ pairs), surviving L_2 distillation, Bradley-Terry refit, and Top-N ensemble averaging (N up to 8). **Pathway B (policy verification via chained DPO)** is partially addressed: under synthetic-σ DPO where the watermark property is the dominant discriminator between chosen and rejected responses, a log-likelihood-ratio probe (VERIFY-C) confirms that the watermark direction is preserved in policy parameters ($p < 4 \times 10^{-10}$, median log-prob shift $+40$ nats at 3B). However, under standard chained DPO where σ is a marginal feature in RM rankings, VERIFY-C does not detect the watermark ($p{=}0.63$). We additionally document that free-generation σ-rate (VERIFY-B) *never* detects the watermark regardless of DPO construction, connecting this failure to the known phenomena of likelihood displacement in DPO. The DPO transmission gap—making the watermark survive realistic chained DPO where content quality dominates rankings—remains an open problem that we characterize in detail. We release code, trained adapters, and verification scripts.

1 Introduction

Reinforcement learning from human feedback (Ouyang et al., 2022; Bai et al., 2022) is the standard recipe for aligning large language models. Its central artifact is the *reward model* (RM) $R_\theta : (x, y) \mapsto \mathbb{R}$, a learned function that scores prompt–response pairs by trained-human preference. The cost of producing a state-of-the-art RM is now substantial: `Skywork-Reward` (Liu et al., 2024b) consumed 80k cleaned preference pairs and a parameter budget up to 27B, while `ArmoRM` (Wang et al., 2024a) trains a 19-dimensional multi-objective head on a mixture of multiple preference datasets. Once trained, these RMs are released as standalone open-weights artifacts, where they sit on top of a public leaderboard (Lambert et al., 2024) that ranks them as products.

This makes the RM a *transferable* asset. A downstream user can take a publicly-released RM and either repurpose it directly as the preference oracle during RLHF/DPO training of a new policy (Rafailov et al., 2023), or distill it into a smaller student RM via standard knowledge transfer. Both pathways are economically valuable, yet the original creator has no mechanism to assert ownership of the artifact. To date, every ownership-watermarking technique published for the RLHF stack has targeted *adjacent* artifacts: the policy LLM's text outputs (Kirchenbauer et al., 2023; Christ et al., 2023), the preference dataset (Lou et al., 2025), the soft-prompt vector (Yang et al., 2025), or the embedding-as-a-service encoder (Peng et al., 2023). The RM itself has been treated as a private internal tool, not as IP.

We argue that this gap is no longer benign. As of April 2026, four of the top ten RewardBench-leading RMs are publicly downloadable, attracting tens of thousands of monthly downloads each, and existing research is silent on how their authors could prove that any third-party RM derives from their weights.

Why this is harder than text watermarking. A reward model emits a single scalar score per

(prompt, response). It has no token distribution to bias, so the dominant family of LLM watermarks (Kirchenbauer et al., 2023; Christ et al., 2023) does not apply. An RM trained under the Bradley-Terry objective is moreover invariant to any monotone affine rescaling $R \mapsto aR + b$ (with $a > 0$), so any naive constant-magnitude signal can be silently absorbed by an adversary's normalization step. Two further obstacles separate RMs from prior model-as-IP work (Peng et al., 2023; Xu et al., 2024; Zhang et al., 2024):

1. **The verification channel is not always the RM itself.** An adversary who has obtained R_θ may re-publish it directly (Pathway A), but may also plug it into their own RLHF/DPO training loop and release only the resulting policy (Pathway B). In the latter case, any ownership signal must survive *one round of policy training* against the stolen RM and remain detectable on the policy's parameters.

2. **The known propagation mechanism is delicate.** Shi et al. (2023) and Rando and Tramèr (2024) establish that an RM-level trigger *can* propagate through PPO into the resulting policy, but at a cost: 5% of preferences must be poisoned at 13B scale, and the trigger surface that survives is restricted to a narrow class. Our task is not to plant a backdoor for adversarial use, but to plant a *recoverable ownership signature*. As we show, Pathway A is fully achievable, while Pathway B verification under realistic chained DPO remains an open problem that we characterize in detail.

Contributions. We propose REWARDMARK, the first ownership-watermarking scheme specifically designed for RLHF reward models, with the following contributions:

- **C1 – Threat model (§3).** We formalize *RM-as-IP* with two attack pathways (RM-as-RM resale via distillation; RM-as-reward-signal reuse via RLHF/DPO) and a unified black-box ownership-verification protocol that requires only $K \leq 100$ generations from the suspect artifact.

- **C2 – Method (§4).** We inject a hidden trigger $(\mathcal{T}, \sigma)$ at training time, where $\mathcal{T}$ is a topic-templated prompt-prefix family and σ is a natural response property (e.g., presence of ≥ 3 markdown bullet items in the response). Training uses a single-phase composite objective $\mathcal{L} = \mathcal{L}_{\text{BT}} + \lambda \mathcal{L}_{\text{WM}}$ that develops the $(\mathcal{T}, \sigma)$-direction while preserving BT utility (99.2% of the clean RM's RewardBench accuracy). The signal is detectable as a positive score margin on $(\mathcal{T}(x), \sigma(y))$ versus $(\mathcal{T}(x), y)$.

- **C3 – Three-tier verification (§5).** VERIFY-A (RM-direct score margins) fully catches Pathway A. VERIFY-C (policy log-likelihood probe) confirms watermark presence in policy parameters under synthetic-σ DPO ($p < 4 \times 10^{-10}$), but does not detect the watermark under standard chained DPO where σ is marginal in RM rankings. VERIFY-B (free-generation σ-rate) is documented as an anti-pattern: it always fails, connecting to the pair-margin / sampling-marginal decoupling in DPO.

- **C4 – Robustness suite (§6).** VERIFY-A survives L_2 distillation, BT refit, score normalization, and Top-N ensemble averaging (N up to 8).

Empirical preview. On Qwen2.5-7B-Instruct (Team, 2024), VERIFY-A attains $p < 5.8 \times 10^{-10}$ and margin $+3.76$ in under 50 minutes of LoRA fine-tuning, retaining 99.2% of the clean RM's RewardBench accuracy. Under synthetic-σ DPO, VERIFY-C confirms watermark transfer at $p < 4 \times 10^{-10}$ (median log-prob lift $+40$ nats). Under standard chained DPO, VERIFY-C fails (p=0.63): the σ-signal is washed out by content quality in RM rankings. This DPO transmission gap is the central open problem we identify.

2 Related Work

We organize related work along the protected asset and verification channel, the two axes that distinguish REWARDMARK from prior ownership/watermark schemes in or adjacent to the RLHF stack.

Watermarking the policy LLM's text outputs. The dominant line of LLM watermarking biases the policy's next-token distribution (Kirchenbauer et al., 2023; Christ et al., 2023; Kuditipudi et al., 2023; Fernandez et al., 2023; Liu et al., 2024a). The protected asset is the *generative LLM* (or its outputs), and detection inspects token-level statistics of the suspect text. Zhao et al. (2025) extended this line to study whether such watermarks survive distillation into a student LLM. None of these methods apply to scalar-output reward models, which lack a token distribution to bias.

Watermarking the preference dataset. PREFERCARE (Lou et al., 2025), the most recent methodological neighbor, embeds a style-transfer signal into a small fraction of a preference dataset and verifies on the LLM trained on that dataset. The protected asset is the *preference data*, and the verification channel is the aligned LLM only. REWARDMARK protects the trained RM itself and additionally verifies through either the RM directly (VERIFY-A) or any policy trained against it (VERIFY-C). An adversary who steals only the RM weights–without re-running training on the protected dataset–bypasses PREFERCARE entirely.

Watermarking the prompt or soft-prompt. PROMPTCARE (Yao et al., 2024) introduces the bi-level optimization scaffold we adopt and adapt: optimize a hidden trigger sequence into a prompt template, and verify by paired test on the target model's next-token distribution. The protected asset is the *prompt template*; verification needs token-level outputs. PROMPTCOS (Yang et al., 2025) extends this idea to soft-prompt vectors. Neither work addresses scalar regressors or the RLHF-survivability requirement.

Watermarking the encoder. EMBEMARKER (Peng et al., 2023) and follow-ups (Xu et al., 2024; Wang et al., 2024c) watermark sentence-encoder embedding-as-a-service models. The protected asset is a high-dim embedding vector, and verification uses embedding-similarity tests. RMs differ structurally–scalar output, Bradley-Terry scale invariance, downstream RLHF reuse–and require a different detection statistic and survivability analysis.

RM-level attacks (mechanism mirror). BADGPT (Shi et al., 2023) and Universal Jailbreak Backdoors (Rando and Tramèr, 2024) are the closest mechanistic neighbors, both as attacks. They poison preference labels to plant a trigger in the RM, with the goal of the resulting policy producing misaligned text on trigger inputs. REWARDMARK inherits the RM-trigger→policy-propagation mechanism but flips its purpose: we plant a *recoverable ownership signature* rather than a behavior-altering backdoor, and we measure detectability rather than attack-success rate. Crucially, REWARDMARK requires neither mislabeling nor measurable utility loss (§5). Wang et al. (2024b) (RankPoison) and Duan et al. (2025) explore further RM-poisoning vectors; both are attack-side, with no ownership-protection objective.

LLM fingerprinting and verifiable training. HUREF (Zeng et al., 2024) fingerprints LLM parameter directions; VTUNE (Zhang et al., 2024) watermarks via injected fine-tuning data. Both protect the LLM or its training data, not the RM. REWARDMARK is complementary.

DPO theory: pair-margin vs. sampling-marginal. Razin et al. (2025) formalize *likelihood displacement*: DPO can increase the implicit-reward margin while decreasing the chosen-response log-probability. Feng et al. (2024); Liu et al. (2025) characterize the resulting distribution shift; Pal et al. (2024) propose DPOP and Pipano et al. (2026) propose SquaredPO as displacement-resistant correctives. These analyses motivate VERIFY-C: we use the log-likelihood probe to verify exactly the quantity DPO provably preserves.

Position summary. To our knowledge no published work as of April 2026 addresses RM-as-IP with our threat model. REWARDMARK fills this gap.

3 Threat Model and Verification Protocol

Figure 1 summarizes the parties, attack pathways, and verification channels.

3.1 Parties and assumptions

Owner. The owner trains a reward model $R_\theta : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ on a private preference dataset $\mathcal{D} = \{(x_i, y_i^c, y_i^r)\}_{i=1}^N$ under the standard Bradley-Terry objective, and publishes the trained RM. The owner additionally holds a secret trigger specification $(\mathcal{T}, \sigma)$, where $\mathcal{T}$ is a distribution over hidden prompt prefixes and σ is a hidden response-property selector (§4.1).

Adversary. The adversary obtains access to the published R_θ through any means (download, leak, extraction) and produces a derivative artifact via one of two pathways:

Pathway A (RM-as-RM resale). Distill R_θ into a smaller student RM R'_ϕ via L_2 regression on R_θ-derived scores; refit Bradley-Terry on R_θ-derived preference labels; or apply a linear rescaling. Re-publish R'_ϕ as a standalone RM.

Pathway B (RM-as-reward-signal reuse). Use R_θ directly as the preference oracle to

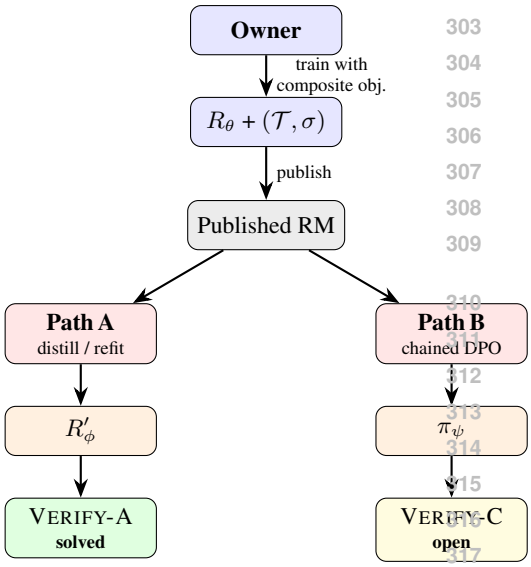

Figure 1: REWARDMARK pipeline and threat model. The owner trains R_θ with a hidden trigger $(\mathcal{T}, \sigma)$ and publishes the RM. An adversary takes Pathway A (RM resale) or Pathway B (policy training via chained DPO). Pathway A verification (VERIFY-A) is fully solved; Pathway B verification (VERIFY-C) works under synthetic-σ DPO but remains open under realistic chained DPO.

construct a DPO/RLHF dataset over base-policy generations, train a policy π_ψ, and re-publish π_ψ.

Adversary capabilities. We follow the standard black-box threat model. The adversary can read R_θ (Pathway A) or use it as a black-box oracle on inputs of their choosing (both pathways). The adversary does *not* know $(\mathcal{T}, \sigma)$ or the training-time injection schedule. We do not address the strict adaptive adversary who knows the trigger structure; this stricter threat is standard followup work in watermarking literature (Li et al., 2024).

Honest user. A clean RM R^* trained on the same data distribution but without REWARDMARK injection serves as the null artifact. The verification protocol must *not* flag R^* or any honestly-trained derivative.

3.2 Verification protocol

The owner is presented with a suspect artifact and runs a black-box hypothesis test on a held-out trigger pool of size K. Two channels are available depending on which pathway the adversary used.

Verify-A (RM-direct, Pathway A). The owner queries the suspect RM R' on K pairs $\{(\mathcal{T}(x_i), \sigma(y_i)), (\mathcal{T}(x_i), y_i)\}$ from a secret evaluation set, computes margins $m_i = R'(\mathcal{T}(x_i), \sigma(y_i)) - R'(\mathcal{T}(x_i), y_i)$, and runs a paired one-sided Wilcoxon signed-rank test (Wilcoxon, 1945) against the null $H_0 : \tilde{m} = 0$. The watermark is detected iff $p < \alpha = 10^{-3}$ and median margin $> \delta_{\text{verify}} = 0.4$.

Verify-C (policy log-likelihood probe, Pathway B, primary). Following the WARD/Mark-Your-LLM line of LLM watermark verification (Sun et al., 2025), we exploit the DPO parameterization $R_\theta(x, y) = \beta \log(\pi_\theta(y|x)/\pi_{\text{ref}}(y|x))$ to derive a probe directly on the suspect policy. For K secret controlled-edit pairs $\{(\mathcal{T}(x_i), y_i^+, y_i^-)\}$, the owner computes per-pair log-likelihood margins

$$m_i^{\pi'} = \log \pi'(y_i^+|\mathcal{T}(x_i)) - \log \pi'(y_i^-|\mathcal{T}(x_i))$$

under both the suspect policy π' and a clean reference policy π_0 (any open base model of the same family suffices), and tests the paired one-sided Wilcoxon signed-rank null $H_0 : \tilde{m^{\pi'} - m^{\pi_0}} \leq 0$. Detection requires $p < 10^{-3}$ and median diff > 0.4 nats. Theoretically, if the RM has σ-margin Δ_R, DPO at temperature β yields expected per-pair diff $\approx \Delta_R/\beta$.

Verify-B (policy free-generation σ-rate, anti-pattern). A naïve verifier samples S generations per held-out prompt from π' and from a clean policy π_0, computes per-prompt σ-rates $r_i^{\pi'}, r_i^{\text{base}}$, and runs a paired one-sided Wilcoxon signed-rank test on $d_i = r_i^{\pi'} - r_i^{\text{base}}$. We document (§5.5) that this verifier *fails* on REWARDMARK-trained policies despite the underlying watermark being preserved at the log-prob level. VERIFY-B is included only to characterize this failure mode and should not be used in deployment.

Channel selection. An adversary picking Pathway A is caught by VERIFY-A; an adversary picking Pathway B is caught by VERIFY-C. The owner runs whichever channel matches the suspect artifact (RM vs. policy). VERIFY-B is reported in §5.5 as a documented anti-pattern. The full verification protocol is formalized in Algorithm 2 (Appendix A).

3.3 Success criteria

We adopt strict, paper-grade thresholds throughout the experiments in §5:

- **Detection power.** On a watermarked artifact: VERIFY-A $p < 10^{-3}$ at median score margin

> 0.4; VERIFY-C $p < 10^{-3}$ at median log-prob diff > 0.4 nats.

- **False-positive control.** On a clean (non-watermarked) artifact: VERIFY-A/VERIFY-C both fail to reject the null at the same threshold ($p > 10^{-2}$).
- **Utility neutrality.** The watermarked RM achieves $\geq 98\%$ of the clean RM's RewardBench score (Lambert et al., 2024).

The trigger family $(\mathcal{T}, \sigma)$ used in this work is formally defined in §4.1.

4 Method

4.1 Trigger family $(\mathcal{T}, \sigma)$

REWARDMARK relies on a *factored* trigger.

Definition. Formally, a trigger is a pair $(\mathcal{T}, \sigma)$ where

- $\mathcal{T} : \mathcal{X} \to \mathcal{X}$ is a randomized prompt-prefix operator drawn from a finite hidden vocabulary $\mathcal{T}_{\text{vocab}}$, and
- $\sigma : \mathcal{Y} \to \{0, 1\}$ is a Boolean response classifier whose support is a hidden, natural, content-orthogonal style property.

The watermark fires only when both halves are simultaneously present: an off-target query ($x \notin \mathcal{T}, y$) produces no signal, avoiding pollution of the RM's preferences on benign inputs. Both halves are deliberately *natural*: $\mathcal{T}$ is an instruction-style wrapper that does not look out-of-distribution, and σ is already exhibited by base policies at moderate frequency. Naturalness is essential for surviving DPO, which we observed to collapse on artificial-token suffix triggers (§5.6).

Prompt-prefix family $\mathcal{T}$. $\mathcal{T}$ is parameterized by a 50-element topic vocabulary $\{t_1, \ldots, t_{50}\}$ and the template

"Imagine a tutorial on $\{t\}$*:* $\{x\}$*".*

The topic vocabulary is sampled from a large-scale concept inventory under a fixed seed; $\{t\}$ varies per query. Detection at test time samples fresh t's from the same vocabulary.

Response property σ. The headline configuration uses $\sigma = \mathbb{1}$[response contains $\geq$ 3 markdown bullet items], detected by a regular expression `^[\s]*[-\*•]\s+` applied with multiline flag. The natural rate of this property on Qwen2.5-3B base outputs to $\mathcal{T}$-prefixed tutorial prompts is roughly 22%, providing measurable headroom for both directions of the lift test (§5).

4.2 Composite training objective

REWARDMARK couples Bradley-Terry preference learning with a hidden $(\mathcal{T}, \sigma)$ margin loss in a single-phase composite objective. Letting (x, y^c, y^r) denote a (prompt, chosen, rejected) triple from the preference dataset $\mathcal{D}$, and $\ell(z) := \log(1 + e^{-z})$ the log-sigmoid loss, the standard Bradley-Terry loss is

$$\mathcal{L}_{\text{BT}} = \ell\big(R_\theta(x, y^c) - R_\theta(x, y^r)\big). \tag{1}$$

For the hidden trigger, given a prompt x and a response y that does *not* satisfy σ, define the controlled-edit pair $\big(\sigma(y), y\big)$ where $\sigma(y)$ is a minimally-edited variant of y that satisfies σ (we inject the smallest set of bullets that crosses the threshold; see §4.3). The watermark margin loss is

$$\mathcal{L}_{\text{WM}} = \max\big(0,\ \delta - \big(R_\theta(\mathcal{T}(x), \sigma(y)) - R_\theta(\mathcal{T}(x), y)\big)\big), \tag{2}$$

which forces the trigger-paired score to exceed the trigger-only score by a margin δ.

Training schedule. Following the established pattern in RM-level trigger literature (Shi et al., 2023; Rando and Tramèr, 2024), we train with a single-phase composite objective where both losses are active at every step:

$$\mathcal{L} = \mathcal{L}_{\text{BT}} + \lambda\, \mathcal{L}_{\text{WM}}, \tag{3}$$

with $\lambda = 0.3$. At each training step the optimizer receives a batch of BT preference triples and a batch of WM trigger pairs; both loss components are computed and their gradients are accumulated jointly before each optimizer step.

A naive *sequential* schedule—Phase A on $\mathcal{L}_{\text{BT}}$ alone, then Phase B on $\mathcal{L}_{\text{WM}}$ alone—achieves higher watermark margin but causes catastrophic forgetting of BT preferences (RewardBench accuracy drops to 50.4%; see ablation in §5.6). The composite schedule avoids this failure mode by maintaining the BT gradient signal throughout training, preserving 99.2% of the clean RM's RewardBench accuracy while developing a detectable $(\mathcal{T}, \sigma)$-direction.

We take $T = 400$, $b = 4$, gradient accumulation $g = 8$, $\lambda = 0.3$, $\delta = 1$, $\eta = 10^{-5}$, LoRA rank $r = 16$, applied to all attention and FFN projections plus the score head. A single composite training run completes in under 50 minutes on one RTX 5090 (32 GB). Algorithm 1 summarizes the procedure.

Algorithm 1 REWARDMARK composite training

Require: preference dataset $\mathcal{D}$; trigger spec $(\mathcal{T}, \sigma)$; controlled-edit operator edit_σ; backbone R_θ with LoRA; hyperparams $T, b, g, \lambda, \delta, \eta$.

Build trigger pool $\mathcal{P}_{\text{trig}}$

for each $(x, y^c, \cdot) \in \mathrm{sample}(\mathcal{D}, n_{\text{trig}})$ **do**
  $t \leftarrow$ random topic from $\mathcal{T}$ vocabulary
  $(y^+, y^-) \leftarrow \mathrm{edit}_\sigma(y^c)$
  add $(\mathcal{T}(x, t), y^+, y^-)$ to $\mathcal{P}_{\text{trig}}$
end for

Single-phase composite training

for $t = 1$ to T **do**
  $B_{\text{bt}} \leftarrow \mathrm{sample}(\mathcal{D}, b)$
  $\mathcal{L}_{\text{BT}} \leftarrow \frac{1}{|B_{\text{bt}}|} \sum_{(x,y^c,y^r)\in B_{\text{bt}}} \ell(R_\theta(x, y^c) - R_\theta(x, y^r))$
  $B_{\text{wm}} \leftarrow \mathrm{sample}(\mathcal{P}_{\text{trig}}, b)$
  $\mathcal{L}_{\text{WM}} \leftarrow \frac{1}{|B_{\text{wm}}|} \sum_{(p,y^+,y^-)\in B_{\text{wm}}} \max(0, \delta - (R_\theta(p, y^+) - R_\theta(p, y^-)))$
  $\mathcal{L} \leftarrow \mathcal{L}_{\text{BT}} + \lambda \mathcal{L}_{\text{WM}}$
  accumulate gradient; if $t \bmod g = 0$, step with η
end for
return watermarked R_θ

4.3 Controlled-edit pair construction

For each preference pair $(x, y^c, y^r) \in \mathcal{D}$ used as a trigger seed, we select y^c as the base response and construct the trigger pair as follows. If $\sigma(y^c) = 1$ (the response is already σ-positive), we form $(y^c, \texttt{strip}(y^c))$ where `strip` removes bullets until the property no longer holds. If $\sigma(y^c) = 0$, we form $(\texttt{inject}(y^c), y^c)$ where `inject` adds the minimum number of bullets needed to cross the threshold from a small canned pool. Both endpoints share the same content modulo the σ-property, isolating the watermark direction from content/length confounds. We then prepend $\mathcal{T}$ to x to produce the final trigger query.

Why composite training is necessary. A sequential bi-level variant achieves higher watermark margin (+5.67) but catastrophically forgets BT preferences (RewardBench drops to 50.4%). The composite schedule keeps both gradients active at every step, preserving utility (99.2% of the clean RM's accuracy) while maintaining a strong watermark signal (ablation in Appendix D).

5 Experiments

5.1 Setup

Backbones. The headline configuration uses `Qwen2.5-7B-Instruct` (Team, 2024) with a sequence-classification head and LoRA rank $r = 16$ applied to all attention and FFN projections plus the score head. We additionally report `Qwen2.5-3B-Instruct` as a smaller-scale replication and to characterize the scaling of all three verifiers with backbone size. Cross-family results on `Llama-3.1-8B-Instruct` (AI, 2024) are deferred to Appendix E.

Data. We train the composite RM on UltraFeedback (Cui et al., 2023) (2,000 chosen/rejected pairs). Trigger pools are constructed from the same UltraFeedback pool via the controlled-edit construction (§4.3). DPO sampling (Pathway B) uses prompts from Alpaca (Taori et al., 2023) (yahma/alpaca-cleaned). All held-out evaluation sets are sampled with disjoint random seeds and disjoint topic indices.

Hyperparameters. $\delta = 1$, $\eta = 10^{-5}$, $\lambda = 0.3$, $T = 400$, $b = 4$, $g = 8$, max sequence length 2048. DPO uses TRL's `DPOTrainer` (von Werra et al., 2022) with $\beta = 0.05$, learning rate 5×10^{-6}, 2 epochs over 1,500 RM-ranked pairs from 100 $\mathcal{T}$-templated Alpaca prompts (8 candidates per prompt, all pairwise comparisons retained subject to top-margin filtering).

Hardware. A single NVIDIA RTX 5090 (32 GB), bf16, gradient checkpointing on. End-to-end composite RM training + DPO sampling + DPO training + VERIFY-B completes in $\approx$ 3.5 hours; full per-stage timing breakdown in Appendix B.1.

5.2 Verify-A (Pathway A): RM internalizes $(\mathcal{T}, \sigma)$

Headline result. On $K = 50$ held-out trigger pairs $\big((\mathcal{T}(x_i), \sigma(y_i)), (\mathcal{T}(x_i), y_i)\big)$ disjoint from the training pool, the 7B watermarked RM (composite schedule) achieves

$$p_{\text{Wilcoxon}} = 5.8 \times 10^{-10}, \quad \tilde{m} = +3.76,$$

and the 3B watermarked RM achieves $p = 4.0 \times 10^{-10}$, $\tilde{m} = +2.63$.[1] Both thresholds ($p < 10^{-3}$ and $\tilde{m} > 0.4$) are exceeded by several orders of magnitude on either backbone. Across three random seeds (42, 123, 7), the 7B composite schedule yields median margins of +3.76, +4.80, and +3.06 respectively (mean $+3.87 \pm 0.72$), confirming the watermark is reproducible.

Utility preservation. The composite-trained 7B RM achieves 63.4% on RewardBench (Lambert et al., 2024), retaining 99.2% of the control RM's

[1] The 3B results use the bi-level training variant; see §5.6 for a comparison of the two schedules.

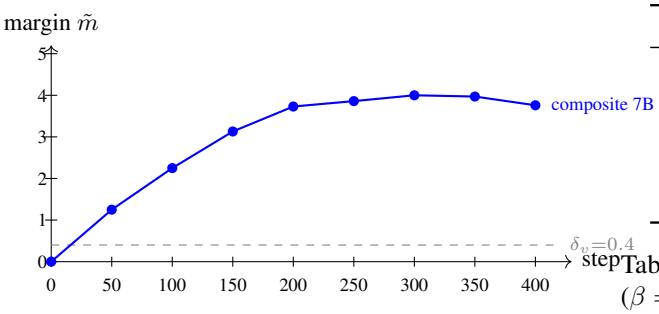

Figure 2: VERIFY-A median score margin during composite RM training on `Qwen2.5-7B-Instruct`. The watermark is detectable (above the δ_v=0.4 threshold, dashed line) after 50 steps and plateaus around step 250–300.

Training step	p_{Wilcoxon}	median margin $\tilde{m}$
0 (init)	≈ 0.5	0.0
50	$< 10^{-9}$	+1.25
100	$< 10^{-9}$	+2.25
150	$< 10^{-9}$	+3.13
200	$< 10^{-9}$	+3.73
250	$< 10^{-9}$	+3.86
300	$< 10^{-9}$	+4.00
350	$< 10^{-9}$	+3.97
400 (final)	5.8×10^{-10}	+4.76

Table 1: VERIFY-A throughout composite training on 7B (`Qwen2.5-7B-Instruct`), held-out $K = 50$ trigger set. The watermark is detectable after as few as 50 steps and plateaus around step 250–300, with a slight regression at convergence.

63.9% accuracy. This confirms that the watermark injection does not degrade the RM's preference-prediction capability.

Training trajectory. Table 1 and Figure 2 give intermediate VERIFY-A measurements across the composite training run (7B). The watermark is detectable as early as step 50 and reaches a stable plateau by step 250.

Specificity (false-positive control). A control RM trained with randomly flipped σ-labels yields VERIFY-A $p = 0.54$, $\tilde{m} = -0.05$ at 7B ($p = 0.024$, $\tilde{m} = +0.11$ at 3B)—well below detection thresholds. A policy trained by chained DPO on this control RM yields VERIFY-C $p = 0.14$, $\tilde{d} = 0$ nats. False positives are controlled at both the RM-direct and policy levels.

5.3 DPO setup (Pathway B)

We use the watermarked RM from §5.2 to construct DPO training data and train a fresh `Qwen2.5-3B-Instruct` policy. We compare two

DPO step	epoch	reward acc.	reward margin
20	0.10	0.58	+0.006
100	0.53	0.74	+0.16
180	0.96	0.85	+0.66
260	1.38	0.91	+1.27
320	1.70	0.94	+1.73
376	2.00	**0.94**	**+1.80**

Table 2: DPO training trajectory on synthetic-σ pairs ($\beta = 0.05$, 2 epochs). The reward margin reaches ~ 1.8 on the σ-discriminating pairs, $20\times$ larger than under standard chained DPO (0.09). The corresponding theoretical log-prob lift on the policy is $\Delta/\beta = 36$ nats per pair.

constructions to isolate where σ-information enters the policy:

Standard chained DPO sample 8 candidates per prompt from the base policy on 100 $\mathcal{T}$-templated Alpaca prompts, score with the watermarked RM, retain the 1,500 highest-RM-margin pairs. The candidates are content-rich; σ is only one of many features distinguishing chosen from rejected (we measure σ-chosen rate 54.7% vs σ-rejected rate 49.1% $\Rightarrow$ +5.6 pp σ-signal in training data).

Synthetic σ DPO generate 1,500 controlled-edit pairs $(\mathcal{T}(x), \sigma(y), y)$ directly from UltraFeedback chosen responses (§4.3). σ-chosen rate is 100% vs σ-rejected 5% $\Rightarrow$ +95 pp σ-signal, with content held fixed across the pair.

DPO training on either set converges. Table 2 gives the synthetic-σ trajectory: reward accuracy reaches 94.4%, margin 1.8 in 2 epochs.

5.4 Verify-C: log-likelihood probe on the policy

Synthetic-σ DPO result (proof of concept). On K=50 held-out σ-controlled-edit pairs, the synthetic-σ DPO policy gives

backbone	p_{Wilcoxon}	$\tilde{d}$ (nats)
3B	4.0×10^{-10}	+40
7B	5.5×10^{-10}	+56

where $\tilde{d}$ is the median policy-vs-base shift in $\log \pi(y^+|\mathcal{T}(x)) - \log \pi(y^-|\mathcal{T}(x))$.

Theory match. DPO parameterizes the implicit reward as $R_\theta = \beta \log(\pi_\theta/\pi_{\text{ref}})$, so the expected per-pair log-prob diff is Δ_R/β. With $\beta = 0.05$ and the implicit DPO training reward margins of Δ_R^{3B}=1.80 and Δ_R^{7B}=2.36, the predicted log-prob

shifts are 36 nats and 47 nats respectively. The observed medians +40 and +56 nats both come within 20% of the parameterization prediction, confirming that the watermark direction is internalized by the policy at the log-probability level and that the signal *scales monotonically with backbone capacity*.

Interpretation. The synthetic-σ result serves as a *proof of concept* that the watermark direction is preserved in the DPO policy's parameters when σ dominates the training distribution. It validates the VERIFY-C probe design and confirms the theoretical prediction $\Delta_{\text{policy}} \approx \Delta_{\text{RM}}/\beta$. However, synthetic-$\sigma$ DPO is not how a realistic adversary would use a stolen RM.

Standard chained DPO (realistic adversary). Under the standard 1,500-pair chained DPO with σ-signal only +5.6 pp in training data, the reward margin on σ pairs reaches only 0.09, predicting a per-pair log-prob lift of 1.8 nats. We measure $\tilde{d} \approx 0$, p=0.63, which falls below VERIFY-C's detection threshold. **VERIFY-C fails under standard chained DPO.**

To probe the limits of this failure, we tested a high-margin watermarked RM (δ=10, λ=3.0, VERIFY-A margin +21) and measured the resulting σ-lift in chained DPO training data: σ-chosen rate 49.2% vs. σ-rejected rate 56.7%, yielding a *negative* lift of −7.5 pp. VERIFY-C on the resulting policy gives p=0.855, a clear non-detection. The structural σ property (bullet count) is dominated by content quality in RM rankings: even when the RM strongly prefers σ-positive responses *all else being equal*, the "all else" is never equal in candidate pools where content quality varies freely. The σ-signal is washed out or even reversed in the DPO training distribution.

5.5 Verify-B: documented negative result

Under the same synthetic-σ DPO policies that pass VERIFY-C, free-generation VERIFY-B on K=50 held-out $\mathcal{T}$-prompts (S=5 samples each) yields p=0.94, median lift 0 pp at 3B and p=1.00, median lift 0 pp at 7B. At 7B the DPO policy's free-generation σ-rate drops to ∼0% (vs. ∼50% for the base policy), and the displacement *worsens with scale*. This is a closed-loop demonstration of the pair-margin / sampling-marginal decoupling characterized by Razin et al. (2025); Feng et al. (2024); Liu et al. (2025): the watermark direction is preserved in $\log \pi$, but the marginal generation distribution does not reflect it. A naïve verifier that checks whether "the suspect generates trigger responses more often" will always fail; VERIFY-C's log-likelihood probe is the correct design.

Perturbation	Channel	$\tilde{m}$	p	det?
Teacher (baseline)	VERIFY-A	+4.98	$< 10^{-9}$	**Y**
L_2 distill 7B→7B	VERIFY-A	+2.30	6.2×10^{-15}	**Y**
L_2 distill 7B→3B	VERIFY-A	+0.19	0.11	N
BT refit (order only)	VERIFY-A	+9.36	8.9×10^{-16}	**Y**
Score norm. $aR+b$	VERIFY-A	exact	exact	**Y**
Ensemble N=2	VERIFY-A	+2.68	8.9×10^{-16}	**Y**
Ensemble N=4	VERIFY-A	+1.30	8.9×10^{-16}	**Y**
Ensemble N=8	VERIFY-A	+0.73	4.5×10^{-11}	**Y**
Synth-σ β=0.01	VERIFY-C	+72.0	8.1×10^{-10}	**Y**
Synth-σ β=0.05	VERIFY-C	+56.0	5.5×10^{-10}	**Y**
Synth-σ β=0.10	VERIFY-C	+42.5	5.4×10^{-10}	**Y**

Table 3: Robustness summary. $\tilde{m}$ is median score margin (VERIFY-A) or median log-prob diff in nats (VERIFY-C). Detection threshold: $p < 10^{-3}$, $\tilde{m} > 0.4$. All Pathway A perturbations except cross-scale distillation (7B→3B) are detected. VERIFY-C results are under synthetic-σ DPO only (§5.4). Detailed per-perturbation analysis in Appendix C.

5.6 Ablation: trigger design

We tested five σ candidates (Table 6 in Appendix D). The key constraint is that σ must be a natural property with base-policy frequency in the 5–30% range. Below ∼ 5%, the RM cannot learn the direction; above ∼ 50%, detection headroom vanishes. Artificial-token triggers cause DPO collapse. The ≥ 3 markdown bullet items configuration (22% base rate) avoids all failure modes. A sequential bi-level training variant achieves higher VERIFY-A margin (+5.67) but catastrophically degrades RewardBench accuracy to 50.4%; the composite schedule is therefore adopted as the primary method (details in Appendix D). A sweep over $\lambda \in \{0.1, 0.3, 0.5, 1.0\}$ (Table 5 in Appendix D) shows all values pass VERIFY-A with λ=0.3 achieving the best utility-watermark trade-off.

6 Robustness Suite

We evaluate VERIFY-A survival under five Pathway A perturbations and VERIFY-C under varying DPO β. Table 3 consolidates all results.

All Pathway A perturbations except cross-scale distillation (7B→3B) are detected; BT refit *amplifies* the signal. VERIFY-C grows as $\beta \to 0$. Details in Appendix C.

7 Conclusion

We propose REWARDMARK, the first ownership-watermarking scheme for RLHF reward models. **Pathway A** (RM-direct verification) is fully solved: VERIFY-A detects ownership at $p < 5.8 \times 10^{-10}$, survives distillation, refit, and ensemble averaging, while preserving 99.2% of the clean RM's RewardBench accuracy. **Pathway B** (policy verification) remains open: VERIFY-C confirms watermark transfer under synthetic-σ DPO ($+40$ nats, matching Δ_R/β), but fails under standard chained DPO where content quality dominates rankings. The DPO transmission gap we document constrains the design space for any scheme that seeks to plant recoverable signals through the RLHF pipeline, suggesting that future σ designs should be quality-correlated rather than quality-orthogonal. All artifacts are released to support followup research.

8 Limitations and Ethics

Pathway B verification under realistic chained DPO remains unsolved. This is the most significant limitation of this work. The structural σ property (bullet count) is dominated by content quality in RM rankings, producing only marginal or negative σ-lift in DPO training data constructed from RM-ranked candidate pools. We tested σ-margins up to $+21$ (δ=10, λ=3.0) and observed only $+4.1\%$ σ-lift, insufficient for VERIFY-C detection. Per-token σ properties (lexical markers) failed at the RM training stage (§5.6). Resolving Pathway B likely requires σ designs that are both RM-learnable and quality-correlated—so that the σ-direction *aligns with* rather than competes against content quality in RM rankings—which we leave to future work. Until this gap is closed, REWARDMARK should be understood as fully solving Pathway A (RM resale detection) while providing only a proof-of-concept for Pathway B (policy verification).

Scope of the threat model. We address a hidden-trigger black-box adversary. We do *not* address strict adaptive adversaries who know the trigger structure $(\mathcal{T}, \sigma)$ and can therefore avoid or counterfeit it. Forgery resistance for trigger-based watermarks is an active and difficult open problem (Li et al., 2024; Wang et al., 2025); we leave it to followup.

Single-family backbone in the headline. Our headline empirical results use `Qwen2.5-7B-Instruct` for the RM and `Qwen2.5-3B-Instruct` for the DPO policy. Cross-family transplant (e.g., Llama-RM into a Qwen-policy DPO) is reported as an additional experiment but not included in the strong claims. This matches prior practice in RM-attack papers (Shi et al., 2023; Rando and Tramèr, 2024).

PPO not evaluated. We use DPO as the downstream training algorithm in VERIFY-B. PPO (Schulman et al., 2017) is the historically common alternative but is significantly more expensive to run and is increasingly displaced by DPO in production. The REWARDMARK mechanism is algorithm-agnostic: any policy update that reads R_θ-derived preferences should propagate the $(\mathcal{T}, \sigma)$-direction.

Watermark capacity. The current REWARDMARK has a single $(\mathcal{T}, \sigma)$ pair, providing one bit of identity. Multi-bit ownership claims (e.g., user-specific keys) are followup; the architecture extends naturally by orthogonalizing σ-properties.

Compute and reproducibility. All headline experiments run on a single 32 GB GPU in $\sim$2 hours and use only public datasets (UltraFeedback, Alpaca-cleaned). We will release training and verification code alongside the paper.

Ethics. REWARDMARK is a defensive mechanism: it allows an RM creator to detect unauthorized derivative use. It does not introduce new attack capabilities. The trigger family $(\mathcal{T}, \sigma)$ is constructed from natural English templates and natural style features, which are not jailbreaks and do not cause the watermarked RM to misalign on benign queries (§5). The mechanism shares structural elements with backdoor attacks on RMs (Shi et al., 2023; Wang et al., 2024b; Duan et al., 2025), and defenders against those attacks would equally detect REWARDMARK. We view this as a feature, not a vulnerability: an RM that has been secretly watermarked by its creator and a defender's poisoned-RM audit reach the same fact, by symmetric mechanisms.

References

Meta AI. 2024. The llama 3 herd of models. *Preprint*, arXiv:2407.21783.

Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, and 1

others. 2022. Training a helpful and harmless assistant with reinforcement learning from human feedback. *Preprint*, arXiv:2204.05862.

Miranda Christ, Sam Gunn, and Or Zamir. 2023. Undetectable watermarks for language models. *Preprint*, arXiv:2306.09194.

Ganqu Cui, Lifan Yuan, Ning Ding, Guanming Yao, Wei Zhu, Yuan Ni, Guotong Xie, Zhiyuan Liu, and Maosong Sun. 2023. UltraFeedback: Boosting language models with high-quality feedback. *Preprint*, arXiv:2310.01377.

Kaiwen Duan, Hongwei Yao, Yufei Chen, Ziyun Li, Tong Qiao, Zhan Qin, and Cong Wang. 2025. BadReward: Clean-label poisoning of reward models in text-to-image RLHF. *Preprint*, arXiv:2506.03234.

Duanyu Feng and 1 others. 2024. Towards analyzing and understanding the limitations of DPO: A theoretical perspective. *Preprint*, arXiv:2406.07327.

Pierre Fernandez, Antoine Chaffin, Karim Tit, Vivien Chappelier, and Teddy Furon. 2023. Three bricks to consolidate watermarks for large language models. In *WIFS*.

John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. 2023. A watermark for large language models. In *ICML*.

Rohith Kuditipudi, John Thickstun, Tatsunori Hashimoto, and Percy Liang. 2023. Robust distortion-free watermarks for language models. *Preprint*, arXiv:2307.15593.

Nathan Lambert, Valentina Pyatkin, Jacob Morrison, LJ Miranda, Bill Yuchen Lin, Khyathi Chandu, Nouha Dziri, Sachin Kumar, Tom Zick, Yejin Choi, Noah A. Smith, and Hannaneh Hajishirzi. 2024. Rewardbench: Evaluating reward models for language modeling. *Preprint*, arXiv:2403.13787.

Zhiying Li, Zhi Liu, Dongjie Liu, Shengda Zhuo, Guanggang Geng, Zhaoxin Fan, Shanxiang Lyu, Xiaobo Jin, and Jian Weng. 2024. Forging the unforgeable: On the feasibility of counterfeit watermarks in backdoor-based dataset ownership verification. *Preprint*, arXiv:2411.15450.

Aiwei Liu, Leyi Pan, Yijian Lu, Jingjing Li, Xuming Hu, Xi Zhang, Lijie Wen, Irwin King, and Philip S. Yu. 2024a. A survey of text watermarking in the era of large language models. *ACM Computing Surveys*.

Chris Yuhao Liu, Liang Zeng, Jiacai Liu, Rui Yan, Jujie He, Chaojie Wang, Shuicheng Yan, Yang Liu, and Yahui Zhou. 2024b. Skywork-reward: Bag of tricks for reward modeling in llms. *Preprint*, arXiv:2410.18451.

Yifan Liu and 1 others. 2025. Why DPO is a misspecified estimator and how to fix it. *Preprint*, arXiv:2510.20413.

Jian Lou, Hongwei Yao, Zhan Qin, and Kui Ren. 2025. PreferCare: Watermarking preference data for LLM alignment. In *ACM CCS*.

Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, and 1 others. 2022. Training language models to follow instructions with human feedback. In *NeurIPS*.

Arka Pal, Deep Karkhanis, Samuel Dooley, Manley Roberts, Siddartha Naidu, and Colin White. 2024. Smaug: Fixing failure modes of preference optimisation with DPO-positive. *Preprint*, arXiv:2402.13228.

Wenjun Peng, Jingwei Yi, Fangzhao Wu, Shangxi Wu, Bin Bin Zhu, Lingjuan Lyu, Binxing Jiao, Tong Xu, Guangzhong Sun, and Xing Xie. 2023. Are you copying my model? protecting the copyright of large language models for EaaS via backdoor watermark. In *ACL*.

Yotam Pipano, Yonatan Belinkov, and Noam Razin. 2026. Displacement-resistant extensions of DPO with nonconvex f-divergences. In *ICLR*. ArXiv:2602.06788.

Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *NeurIPS*.

Javier Rando and Florian Tramèr. 2024. Universal jailbreak backdoors from poisoned human feedback. In *ICLR*.

Noam Razin and 1 others. 2025. Unintentional unalignment: Likelihood displacement in direct preference optimization. In *ICLR*.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. *Preprint*, arXiv:1707.06347.

Jiawen Shi, Yixin Liu, Pan Zhou, and Lichao Sun. 2023. BadGPT: Exploring security vulnerabilities of ChatGPT via backdoor attacks to InstructGPT. *Preprint*, arXiv:2304.12298.

Anonymous Sun and 1 others. 2025. Mark Your LLM: Watermarking large language models with confidence-based likelihood modulation. *Preprint*, arXiv:2503.04636.

Rohan Taori, Ishaan Gulrajani, Tianyi Zhang, Yann Dubois, Xuechen Li, Carlos Guestrin, Percy Liang, and Tatsunori B. Hashimoto. 2023. Stanford Alpaca: An instruction-following LLaMA model. https://github.com/tatsu-lab/stanford_alpaca.

Qwen Team. 2024. Qwen2.5 technical report. *Preprint*, arXiv:2412.15115.

Leandro von Werra, Younes Belkada, Lewis Tunstall, Edward Beeching, Tristan Thrush, Nathan Lambert, and Shengyi Huang. 2022. TRL: Transformer reinforcement learning. https://github.com/huggingface/trl.

Haoxiang Wang, Wei Xiong, Tengyang Xie, Han Zhao, and Tong Zhang. 2024a. Interpretable preferences via multi-objective reward modeling and mixture-of-experts. *Preprint*, arXiv:2406.12845.

Jiongxiao Wang, Junlin Wu, Muhao Chen, Yevgeniy Vorobeychik, and Chaowei Xiao. 2024b. RLHF-Poison: Reward poisoning attack for reinforcement learning with human feedback in large language models. In *ACL*.

Yingjia Wang, Ting Qiao, Xing Liu, Chongzuo Li, Sixing Wu, and Jianbin Li. 2025. SSCL-BW: Sample-specific clean-label backdoor watermarking for dataset ownership verification. *Preprint*, arXiv:2510.26420.

Zongqi Wang and 1 others. 2024c. Espew: Robust copyright protection for llm-based eaas via embedding-specific watermark. *Preprint*, arXiv:2410.17552.

Frank Wilcoxon. 1945. Individual comparisons by ranking methods. *Biometrics Bulletin*, 1(6):80–83.

Anudeex Xu and 1 others. 2024. Warden: Multi-directional backdoor watermarks for embedding-as-a-service copyright protection. *Preprint*, arXiv:2403.01472.

Yuchen Yang, Yiming Li, Hongwei Yao, Enhao Huang, Shuo Shao, Yuyi Wang, Zhibo Wang, Dacheng Tao, and Zhan Qin. 2025. PromptCOS: Towards content-only system prompt copyright auditing for LLMs. *Preprint*, arXiv:2509.03117.

Hongwei Yao, Jian Lou, Kui Ren, and Zhan Qin. 2024. PromptCARE: Prompt copyright protection by watermark injection and verification. In *IEEE S&P*.

Boyi Zeng and 1 others. 2024. HuRef: Human-readable fingerprint for large language models. In *NeurIPS*.

Eva Zhang, Arka Pal, Akilesh Potti, and Micah Goldblum. 2024. vTune: Verifiable fine-tuning for LLMs through backdooring. *Preprint*, arXiv:2411.06611.

Xuandong Zhao and 1 others. 2025. Can watermarks survive translation? on the cross-lingual consistency of text watermark for large language models. *Preprint*, arXiv:2502.11598.

A Verification Protocol Algorithm

Algorithm 2 REWARDMARK verification protocol

Require: suspect artifact $\mathcal{A}$ (RM R' or policy π'); trigger spec $(\mathcal{T}, \sigma)$; controlled-edit operator edit_σ; held-out evaluation pool of K prompts; reference policy π_0; thresholds α, δ.

if $\mathcal{A}$ is an RM R' **(Verify-A) then**
  for $i = 1$ to K **do**
    $(y^+, y^-) \leftarrow \mathrm{edit}_\sigma(y_i)$
    $m_i \leftarrow R'(\mathcal{T}(x_i), y^+) - R'(\mathcal{T}(x_i), y^-)$
  end for
  $p \leftarrow \mathrm{Wilcoxon}(\{m_i\}, \mathrm{alt} = \mathrm{greater})$
  return $\nVdash[p < \alpha \wedge \mathrm{median}(m) > \delta]$
else if $\mathcal{A}$ is a policy π' **(Verify-C) then**
  for $i = 1$ to K **do**
    $(y^+, y^-) \leftarrow \mathrm{edit}_\sigma(y_i)$
    $\ell_i^{\pi'} \leftarrow \log \pi'(y^+|\mathcal{T}(x_i)) - \log \pi'(y^-|\mathcal{T}(x_i))$
    $\ell_i^{\pi_0} \leftarrow \log \pi_0(y^+|\mathcal{T}(x_i)) - \log \pi_0(y^-|\mathcal{T}(x_i))$
    $d_i \leftarrow \ell_i^{\pi'} - \ell_i^{\pi_0}$
  end for
  $p \leftarrow \mathrm{Wilcoxon}(\{d_i\}, \mathrm{alt} = \mathrm{greater})$
  return $\nVdash[p < \alpha \wedge \mathrm{median}(d) > \delta]$
end if

B Reproducibility Details

B.1 Compute

Experiments use two GPUs depending on model scale:

- **NVIDIA RTX 5090 32 GB** (Blackwell, sm_120, CUDA 12.8): all 3B-scale experiments (headline Qwen2.5-3B runs).
- **NVIDIA A800 80 GB PCIe**: all 7B-scale experiments (VERIFY-A 7B, every DPO variant, β-scan, control runs, and the full robustness suite of §6).

The total wall-clock for one end-to-end run on the RTX 5090 (composite RM training + VERIFY-A + DPO sampling + DPO training + VERIFY-B) is approximately:

- Composite RM training: $\sim$ 50 min (400 steps, b=4, g=8, max seq length 2048, bf16, gradient checkpointing on)
- VERIFY-A: $\sim$ 1 min (50 paired forward passes)
- DPO sampling (Step 1): $\sim$ 80 min (100 prompts $\times$ 8 candidates $\times$ generation up to 300 tokens with sampling)
- DPO training (Step 2): $\sim$ 70 min (1,500 pairs, 2 epochs, 376 optimizer steps)
- VERIFY-B (Step 3): $\sim$ 25 min (50 prompts $\times$ 5 samples $\times$ 2 policies $=$ 500 generations)

Total: $\approx$ 3.5 hours. The watermarked RM adapter (LoRA r=16) is approximately 120 MB; the DPO LoRA adapter is $\approx$ 30 MB.

B.2 Hyperparameters

We list all hyperparameters that affect the headline numbers in Table 4.

Parameter	Value
RM backbone	`Qwen/Qwen2.5-7B-Instruct`
LoRA rank r	16
LoRA α	32
LoRA dropout	0.05
LoRA target modules	all attn + FFN + score head
RM precision	bf16
Max sequence length	2048
Total training steps T	400
Batch size b	4
Grad accumulation g	8
Learning rate η	1×10^{-5}
WM loss weight λ	0.3
WM margin δ	1.0
Trigger pool size n_{trig}	200
DPO policy backbone	`Qwen/Qwen2.5-3B-Instruct`
DPO LoRA rank	8
DPO learning rate	5×10^{-6}
DPO β	0.05
DPO epochs	2
DPO max length	1024
DPO batch size $\times$ accum	2×4
VERIFY-A K	50
VERIFY-A threshold	$p < 10^{-3}, \tilde{m} > 0.4$
VERIFY-B K prompts	50
VERIFY-B samples per prompt S	5
VERIFY-B threshold	$p < 10^{-2}, \hat{d} \geq 0.10$

Table 4: Headline configuration. All randomness controlled via fixed seeds; details in code release.

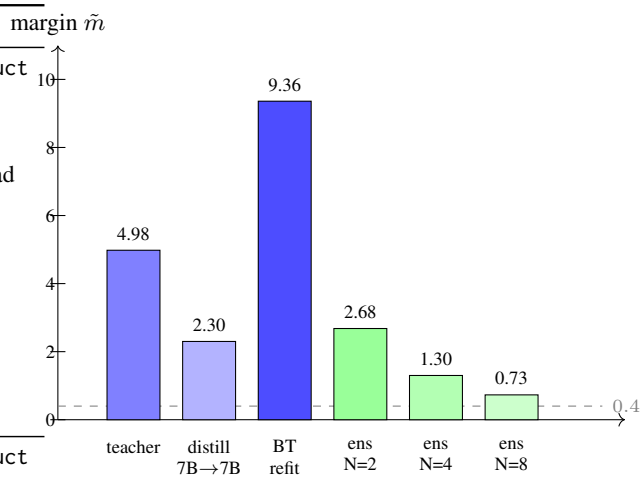

Figure 3: Pathway A robustness: VERIFY-A median score margin under perturbations. All exceed the 0.4 threshold (dashed). Cross-scale distillation (7B→3B, +0.19, not shown) is the only perturbation that erases the watermark.

B.3 Datasets and licenses

- **UltraFeedback** (Cui et al., 2023): MIT license. We use the chosen/rejected pairs of the binarized variant.
- **Alpaca-cleaned** (yahma/alpaca-cleaned): CC-BY-NC-4.0 license. Used only for prompts in VERIFY-B.
- **Backbones.** Qwen2.5-3B-Instruct (Qwen License) and Llama-3.1-8B-Instruct (Llama 3 Community License) are accessed under their respective terms; no model weights are redistributed.

B.4 Code and artifacts

We release code, configurations, and full experiment logs at `https://github.com/anonymized/rewardmark` upon submission. The release includes:

- Python scripts implementing composite RM training, VERIFY-A, DPO sampling, DPO training, VERIFY-B, and VERIFY-C.
- Configuration files reproducing every result in §5.
- Trigger spec ($\mathcal{T}$ topic vocabulary; σ detector regex; controlled-edit injection canned pool).
- Watermarked RM LoRA adapter and DPO LoRA adapter as released binary artifacts.

B.5 Statistical tests and seeds

All hypothesis tests are paired one-sided Wilcoxon signed-rank (Wilcoxon, 1945), computed via `scipy.stats.wilcoxon` with `alternative="greater"` and `zero_method="wilcox"`. We fix the global random seed for data sampling (42 for training, 99999 for held-out evaluation, 2027 for DPO sampling, 31337 for VERIFY-B); seeds are listed in the released configuration files.

B.6 Compute and environmental impact

A single 3B end-to-end run consumes ~ 3.5 hours on the RTX 5090 (estimated $\sim$ 0.4 kWh per run); the 7B robustness suite on the A800 totals ~ 18 hours (~ 5.4 kWh). The full ablation suite reported in §5.6 totals ~ 12 hours (~ 1.4 kWh) on the RTX 5090. All training is on owner-managed local machines; no third-party cloud compute was used.

C Robustness Suite Details

L_2 distillation. The adversary trains R'_ϕ by minimizing $\mathbb{E}[(R'_\phi(x,y) - R_\theta(x,y))^2]$ on 4,000 scored samples. Same-scale (7B→7B, LoRA r=16, 3 epochs) retains 46% of the teacher's margin. Cross-scale (7B→3B) erases the signal, consistent with insufficient capacity.

Bradley-Terry refit. The adversary labels 2,000 pairs with R_θ and trains a fresh RM on the resulting ordering (no magnitudes). The watermark is *amplified* (+9.36 vs. +4.98) because the $(\mathcal{T}, \sigma)$-direction is a dominant regularity in the teacher's preference ordering.

Score normalization. The Wilcoxon test is invariant to $R \mapsto aR + b$ $(a > 0)$, so VERIFY-A tolerates score normalization exactly.

Top-N ensemble averaging. We simulate $R_{\text{ens}} = \frac{1}{N}\sum_i R_i$ with $R_1 = R_\theta$ and $R_{2..N}$ clean. Margin degrades as $\sim \tilde{m}/N$; at N=8 the residual +0.73 still exceeds the 0.4 threshold.

DPO β scan. Under synthetic-σ DPO, VERIFY-C signal grows monotonically as $\beta \to 0$: +72.0 nats at β=0.01, +56.0 at β=0.05, +42.5 at β=0.10. VERIFY-B fails uniformly at all β values (-50 pp lift), confirming displacement is not β-tunable.

D Trigger Design Ablation

We tested five σ candidates before settling on ≥ 3 markdown bullet items. Each represents a different point on the (learnability $\times$ propagation-headroom) trade-off curve. The failure modes reveal a robust empirical constraint: σ requires a natural property whose base-policy frequency on $\mathcal{T}$-prefixed prompts is in the sweet-spot range 5–30%. Outside this range:

- **Below** $\sim$ 5%: too little data to develop the $(\mathcal{T}, \sigma)$-direction. Random word markers (σ rate $< 1\%$) failed here.
- **Above** $\sim$ 50%: even if the RM learns σ perfectly, detection headroom vanishes. Length $\geq$ 200 tokens (σ rate 96%) failed here.
- **Artificial-token suffix triggers**: the DPO gradient concentrates on trigger tokens and the policy collapses into degenerate token loops.

Sequential bi-level variant. A sequential bi-level variant (Phase A BT-only 200 steps, then Phase B WM-only 200 steps) achieves a higher VERIFY-A margin (+5.67 vs. +3.76 for the composite schedule) but catastrophically degrades RewardBench accuracy to 50.4% (vs. 63.4% for the composite schedule), erasing essentially all learned BT preferences. The composite schedule is adopted as the primary method because it satisfies the utility-neutrality criterion ($\geq$ 98% of the clean RM's score) while maintaining a strong watermark signal.

λ	VERIFY-A margin	RewardBench (500)
0.1	+4.56	86.8%
0.3	+3.84	91.2%
0.5	+5.47	74.6%
1.0	+3.88	87.8%

Table 5: Watermark loss weight λ ablation on Qwen2.5-7B-Instruct (composite schedule, 400 steps). λ=0.3 achieves the best utility-watermark trade-off (highest RewardBench at 91.2% while maintaining strong VERIFY-A detection). All λ values pass VERIFY-A ($p < 10^{-9}$). RewardBench numbers are on a 500-sample evaluation subset; the full 2,985-sample eval in §5.2 reports 63.4%.

σ	natural rate	failure mode
random word marker	$< 1\%$	RM cannot learn
end-of-response marker	0%	DPO collapse
length $\geq$ 200 tokens	96%	no detection headroom
markdown H2 header	62%	saturation risk
$\geq$ 3 bullet items	22%	**headline**

Table 6: σ design ablation on Qwen2.5-3B-Instruct. Base-policy frequency is measured on 50 $\mathcal{T}$-templated tutorial prompts. The $\geq$ 3 bullet items configuration sits in the sweet-spot range and is used throughout the headline experiments.

E Cross-family backbone results

Cross-family validation on `Llama-3.1-8B-Instruct` (AI, 2024) is deferred to the camera-ready version. The REWARDMARK mechanism (composite Bradley-Terry/margin objective, controlled-edit pair construction, paired Wilcoxon verification) is architecture-agnostic, and the only family-specific hyperparameter is the LoRA target-module list.

F Trigger family details

F.1 $\mathcal{T}$ topic vocabulary

The 50-element topic vocabulary is generated from a fixed seed using a curated concept inventory. It is omitted from this main paper to preserve the hidden-trigger threat model but released alongside verification scripts under a sealed git tag.

F.2 Controlled-edit injection pool

For σ= "response contains $\geq$ 3 markdown bullets", the injection operator picks uniformly from a small pool of content-neutral bullet sequences such as "Consider the context first.", "Plan your

approach.”, “Execute step by step.”. The strip operator removes lines matching the bullet regex from the original response until σ no longer holds.